



**UNIVERSITY ENTRANCE TEST FOR
ADMISSION TO UNDERGRADUATE PROGRAMMES - 2025**

Booklet No. :

29666

Series :

B

Time Allowed: 3 Hours (180 Minutes).

This booklet contains 180 questions and 24 printed pages.
Do not open this booklet until you are asked by the Invigilator.

Correct way to response a question on OMR sheet

Correct way to Answer



Incorrect way



Important Note:

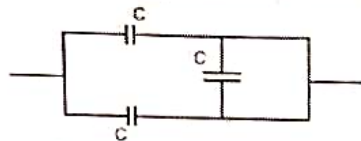
There are 180 MCQ's under Physics, Chemistry, Biology and Mathematics with 45 questions in each section.

Instructions to be followed:

1. Immediately after opening the Question Booklet, ensure that it contains the number of pages and questions as depicted above. In case of any discrepancy, get the question booklet replaced with the question booklet of same series.
2. Use only blue/black ball point pen for writing your particulars like Roll No., Question Booklet No., Question Booklet series and for marking answers on the OMR answer sheet at the spaces provided. OMR answer sheets without mentioning Question Booklet No., and Question Booklet Series will not be evaluated.
3. Do not write anything else on OMR answer sheet. Mark only the responses to the questions on OMR Sheet. For rough work use the blank spaces provided on question booklet.
4. The candidate are not allowed to use white correction fluid on the OMR Answer Sheet and Attendance Sheet.
5. Erasing, cutting, overwriting on the OMR sheet in answering circles and Marking multiple answers for a question will be deemed as wrong answer and no mark will be awarded for such answers.
6. All Questions carry equal point of ONE (1) for correct answer, and there is **no negative marking**.
7. The candidates should on demand show their admit cards to the invigilators or any other authorized representative of SKUAST-K.
8. No candidate will be allowed to leave the examination hall until the examination time is over.
9. The candidates shall handover the OMR Answer Sheet (Original) to the invigilator before leaving the examination hall.
10. The candidate is not allowed to carry any stationary items Printed / Written, Bits of Paper, Geometry / Pencil Box, Plastic Pouch, Calculator, Writing Pad, Pen Drives, Eraser, Log Table, Electronic Pen / Scanner etc.
11. The Candidate is NOT allowed to carry any communication device like Mobile Phone, Bluetooth Earphones, Microphones, Health Band, Smart Watch, Camera & other items like Wallet, Sunglasses, Handbags, Belt, Cap, ATM / Credit / Debit Cards, any remote key etc.
12. The candidate should sit on their allotted seats only.
13. Any act of indiscipline/misconduct/use of unfair means shall result in forfeiture of candidate's right to sit in the examination. The decision of Chief Observer / Controller of Examinations and Superintendent of the Centre in this regard shall be the final.
14. For all subsequent purposes your Roll No. shall remain same as provided in the admit card.

- Q 8. Two thin spherical shells made of metal are at a large distance apart; one of radius 10 cm carries a charge of $0.5 \mu\text{C}$, and the other of radius 20 cm carries a charge of $0.7 \mu\text{C}$. The charge on each when a conducting wire connects them is respectively
- A 0.4 and $0.8 \mu\text{C}$
 B 0.5 and $0.7 \mu\text{C}$
 C 0.425 and $0.85 \mu\text{C}$
 D 0.6 and $0.6 \mu\text{C}$
- Q 9. A circular loop of radius R carrying a current I is placed in a uniform magnetic field with its plane perpendicular to B . The torque on the loop is -
- A $\pi R^2 I B$
 B $\pi R^2 I^2 B$
 C $\pi I R^2 B^2$
 D Zero
- Q 10. The distance between the objective lens and the eye lens of an astronomical telescope when adjusted for parallel light is 100 cm. The measured value of the magnification is 19. The focal length of the lens is
- A 85 & 15 cm
 B 82 & 18 cm
 C 95 & 5 cm
 D 50 & 50 cm

- Q 11. The equivalent capacitance of the combination shown in the figure is:

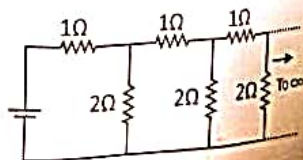


- A $\frac{3C}{2}$
 B $3C$
 C $2C$
 D $\frac{C}{2}$

- Q 12. A power transmission line feeds input power at 2300V to step-down transformer with its primary windings having 4000 turns. The power output is delivered at 230V by the transformer. If the current in the primary of the transformer is 5A and its efficiency is 90%, the output current would be -
- A 25A
 B 50A
 C 35A
 D 45A

- Q 13. The second-order bright fringe in a single slit diffraction pattern is 1.4 mm from the centre. The screen is 80 cm from the slit of width 0.80 mm. Assuming monochromatic incident light, the wavelength of light will be -
- A $5800 \text{ \AA}$
 B $5600 \text{ \AA}$
 C $8000 \text{ \AA}$
 D $1400 \text{ \AA}$

- Q 14. As shown in fig, an infinite ladder network of resistance is connected with 1Ω and 2Ω . The effective resistance between A and B is -



- A 4Ω
 B 3Ω
 C 2Ω
 D Infinite

- Q 15. Considering the junction diode as ideal, the value of current flowing through AB is -



- A 0
 B 10^{-2} A
 C 10^{-3} A
 D 10^{-3} A

- Q 16. For a spherical surface of radius of curvature R , separating two media of refractive indices μ_1 and μ_2 the two principal focal lengths are f_1 and f_2 respectively. Which one of the following relations is correct?

A $f_1 = f_2$

B $\frac{f_2}{\mu_2} = \frac{f_1}{\mu_1}$

D $\frac{f_2}{\mu_1} = \frac{f_1}{\mu_2}$

$\frac{f_2}{\mu_2} = -\frac{f_1}{\mu_1}$

$R = \frac{\mu_1 \mu_2}{f_1 f_2}$
 $R = \frac{\mu_1 \mu_2}{f_1 f_2}$

- Q 17. Two circular coils can be arranged in any of the three situations shown below. The mutual inductance will be -

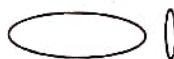
(a)



(b)



(c)



- A Maximum in (a)
C Maximum in (c)

- B Maximum in (b)
D Same in all situation

- Q 18. The relative permittivity of distilled water is 81. The velocity of light in it will be - (Take $\mu_r = 1$)

A 4.33×10^7 m/s

C 3.33×10^7 m/s

B 2.33×10^7 m/s

D 5.33×10^7 m/s

- Q 19. Pick the wrong answer in context with rainbow -

A An observer can see a rainbow when his front is towards the Sun.

C When the light rays undergo two internal reflections in a water drop, a secondary rainbow is formed.

B A rainbow is a combined effect of dispersion, refraction, and reflection at light.
D The order of colours is reversed in the secondary rainbow.

- Q 20. In Young's double slit experiment, the ratio of the slit widths is 4:1. The ratio of the intensity of maxima to minima close to the central fringe on the screen will be -

A 25:9

C $(\sqrt{3} + 1)^4 : 16$

B 4:1

D 9:1

- Q 21. Two electric bulbs rated P_1 and P_2 watt at V volt are connected in series across V volt mains, then total power consumption P is -

A $P_1 + P_2$

C $\frac{P_1 P_2}{P_1 + P_2}$

B $\sqrt{P_1 P_2}$

D $\frac{P_1 + P_2}{P_1 P_2}$

- Q 22. The frequency of the incident light falling on a photosensitive metal plate is doubled. The Kinetic energy of the emitted photo electrons is -

A Double of the earlier value

C More than double

B Unchanged

D Less than double

- Q 23. An air bubble in a glass slab ($\mu = 1.5$), when viewed from one side appears to be at 6 cm and from the opposite side 4 cm. The thickness of the glass slab is -

A 10 cm

C 15 cm

B 6.67 cm

D None

- Q 24. A cylinder of radius R and L is placed in a uniform electric field E parallel to the cylinder axis. The total flux for the surface of the cylinder is given by -

A $2\pi R^2 E$

C $3\pi R^2 E$

B $2\pi R^2 E$

D Zero

- Q 25. A given length of wire carries a steady current. It is bent to form a circular plane coil of one turn. If a loop of same length is now bent more sharply to give a double loop of smaller radius, the magnetic field at the centre caused by the same current is -
- A A quarter of its first value
B unaltered
C Four times of its first value
D Half of its first value

- Q 26. A vessel is of depth $2h$. Its lower half is filled with a liquid of refractive index $2\sqrt{2}$ and upper half with another liquid of R.I. $\sqrt{2}$. The liquids are immiscible. The apparent depth of the inner surface of the bottom of the vessel will be -

- A $\frac{h}{\sqrt{2}}$
B $\frac{3}{4}h\sqrt{2}$
C $\frac{h}{2(\sqrt{2} + 1)}$
D $\frac{3h}{2\sqrt{2}}$

- Q 27. Which of the following transitions in hydrogen atom emit photons of higher frequency -

- A $n = 1$ to $n = 2$
B $n = 2$ to $n = 6$
C $n = 6$ to $n = 2$
D $n = 2$ to $n = 1$

- Q 28. A long cylindrical volume contains a uniformly distributed charge of density ρ . The radius of cylindrical volume is R . A charge particle q revolves around the cylinder in a circular path. The kinetic energy of the particle is -

- A $\frac{\rho q R^2}{4\epsilon_0}$
B $\frac{\rho q R^2}{2\epsilon_0}$
C $\frac{\rho q}{4\epsilon_0 R^2}$
D $\frac{4\epsilon_0 R^2}{\rho q}$

- Q 29. Twenty-seven drops of the same size are charged at 220V each. They combine to form a bigger drop. The potential of the bigger drop is -

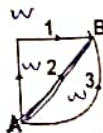
- A 1980V
B 660V
C 1320V
D 1520V

- Q 30. Two capacitors, each having a capacitance $40\mu\text{F}$, are connected in series. The space between one of the capacitors is filled with dielectric material of dielectric constant K such that the equivalent capacitance of the system becomes $24\mu\text{F}$. The value of K will be -

- A 1.5
B 2.5
C 1.2
D 3

- Q 31. If W_1 , W_2 and W_3 represent the work done in moving a particle from A to B along three different paths 1, 2 and 3 respectively (as shown in figure) in the gravitational field of a point mass m the correct relation between W_1 , W_2 and W_3 will be -

- A $W_1 > W_2 > W_3$
B $W_1 = W_2 = W_3$
C $W_1 < W_2 < W_3$
D $W_2 > W_1 > W_3$



$W_1 = W_2 = W_3$

- Q 32. Two sound sources produce progressive vibrations $Y_1 = 4 \sin 400 \pi t$ and $Y_2 = 3 \sin 404 \pi t$ are placed close to the ear of a person. He hears -

- A 2 beats/sec and the ratio of maximum and minimum intensities is 4/3
B 2 beats/second and the ratio of maximum and minimum intensities is 49/1
C 4 beats/second and the ratio of maximum and minimum intensities is 49/1
D 4 beats/second and the ratio of maximum and minimum intensities is 7/1

$4 \sin 400 \pi t$
 $3 \sin 404 \pi t$
 $y_2 = 3 \sin 404 \pi t$

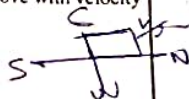
- Q 33. A particle of mass m moving eastward with a speed v collides with the same mass moving northward with the same speed v . The two particles coalesce after collision. The combined body will move with velocity -

A v in north east

B $\frac{v}{\sqrt{2}}$ in south west

☒ C $\frac{v}{\sqrt{2}}$ north east

D $v\sqrt{2}$ south east



- Q 34. A particle of mass 3 Kg is moving under the action of a central force whose potential energy is given by $U = 10 r^3$ joule. For what energy will the orbit be a circle of radius 10 m?

A 2.5×10^4 J

B 1.25×10^4 J

☒ C 1.5×10^4 J

D Zero

- Q 35. A simple pendulum performs SHM about $x = 0$ with an amplitude A and time period T . The speed of the pendulum at $x = A/2$ will be -

A $\frac{A}{T}$

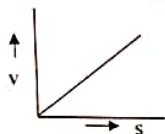
B $\frac{\sqrt{3}}{T} \pi A$

☒ C $\frac{\sqrt{3}}{2T} \pi A$

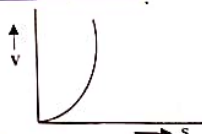
D $\frac{3\pi^2 A}{T}$

- Q 36. An object is dropped from rest. Its velocity versus displacement graph is -

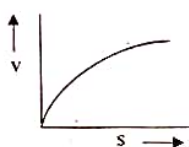
A



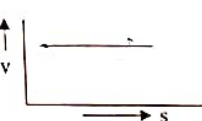
B



C



☒ D



- Q 37. In a large room, a person receives direct sound waves from a source 120 m away from him. He also receives waves from the same source which reach him, being reflected from the 25 m high ceiling at a point half way between them. For which of the following set of wavelengths, will these interfere constructively?

A 10 m, 5 m, 10/3 m

☒ B 25 m, 8 m, 17 m

C 12 m, 5 m, 13 m

D 10 m, 2 m, 1/3 m

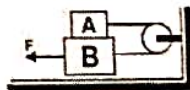
- Q 38. Block A weighs 4 N and block B weighs 8 N. The coefficient of static friction is 0.25. The value of force F to slide B at a constant speed when A & B are connected by a light cord passing over a smooth pulley is -

A 5 N

B 8 N

C 4 N

☒ D 12 N



- Q 39. $\bar{v}$, v_{rms} and v_p are the mean speed, root mean square speed and most probable speed of the molecules in an ideal monoatomic gas at absolute temperature T . The mass of the molecule is m . Then -

A No molecule can have speed greater than

B No molecule can have speed less than $\frac{v_p}{\sqrt{2}}$

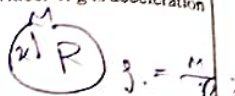
$\sqrt{2} v_{rms}$

C $v_p < \bar{v} < v_{rms}$

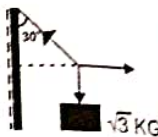
☒ D The average Kinetic Energy of a molecule is $\frac{5}{2} m v_p^2$

- Q 40. Which of the following statements is not correct about the soap bubble -
 A Work done in forming the bubble of radius R is $4\pi R^2 T$.
 B Work done in doubling the radius of a bubble of radius R and surface tension T is $24\pi R^2 T$.
 C Pressure inside the bubble is greater than inside a drop of same radius and of same liquid.
 D Pressure inside the bubble is more than outside it.

- Q 41. A satellite of mass m revolves around the earth of radius R at a height x from its surface. If g is acceleration due to gravity on the surface of the earth, the orbital speed of the satellite is -
 A gx
 B $\frac{gR}{R-x}$
 C $\frac{gR^2}{R+x}$
 D $\sqrt{\frac{gR^2}{R+x}}$



- Q 42. A block of mass $\sqrt{3}$ kg is attached to a string whose other end is attached to a wall. An unknown force F is applied so that the string makes an angle of 30° with the wall. The tension T is -
 ($g = 10 \text{ m/s}^2$)



A 20N
 C 10N

B 25N
 D 15N

- Q 43. Which of the following measurements is most precise?

A 5.00 mm
 C 5.00 m

B 5.00 cm
 D 5.00 Km

- Q 44. The density of a non-uniform rod of length 1 m is given by $\rho(x) = a(1 + bx^2)$, where a and b are constants and $0 \leq x \leq 1$. The centre of mass of the rod will be at -

A $\frac{3(2+b)}{4(3+b)}$

B $\frac{4(2+b)}{3(3+b)}$

C $\frac{3(3+b)}{4(2+b)}$

D $\frac{4(3+b)}{3(2+b)}$

1m

- Q 45. A merry-go-round made of a ring like a platform of radius R and mass M is revolving with angular speed ω . A person of mass M is standing on it. At one instant, person jumps off the round radially away from the centre of the round. The speed of the round afterwards will be -

A Ω
 C $\omega/2$

B 2ω
 D 0

Chemistry (Q. No. 46 to Q. No. 90.)
 (For Candidates Opted for PCB / PCM / PCBM Stream)

- Q 46. The tert-butyl carbocation has which type of C-C bonds?

A sp^3-sp^3

B sp^3-p
 C sp^3-sp^3

- Q 47. When the solid and liquid states of a substance reach a state of equilibrium at a given pressure and temperature, then which one of the following observation(s) is correct?

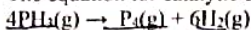
A Mass of both solid and liquid remains constant
 B Temperature of the system remains constant.

C Both A and B

D Neither A nor B

- Q 48. Petrol is a mixture of hydrocarbons that is used for removal of grease stains during dry cleaning. What is the nature of grease?
 A Polar ~~B Non-polar~~
 C Both A and B ~~D Neither A nor B~~
- Q 49. Benzene, when treated with *n*-butyl chloride in presence of anhydrous aluminium chloride, gives which one of the following as the major product?
 A *n*-Butyl benzene ~~B *Sec*-butyl benzene~~
~~C *tert*-Butyl benzene~~ ~~D Isobutyl benzene~~
- Q 50. Which of the following will be formed when 1,3,5-trimethyl benzene undergoes ozonolysis?
 A CH_3COCHO ~~B CH_3CHO~~
~~C CH_3COOH~~ ~~D $\text{CH}_3\text{CHO} + \text{HCHO}$~~ 1, 3, 5
- Q 51. The humidity in air is classified as which type of solution?
 A Gas in gas ~~B Liquid in liquid~~
~~C Gas in liquid~~ ~~D Liquid in gas~~
- Q 52. How much is 100 ppm in g/mL.
 A 0.1 ~~B 0.01~~
~~C 0.001~~ ~~D 0.0001~~
- Q 53. For a solution of two liquids, choose the correct statement(s) from the following:
 (i) Solutions having strong solute-solvent interactions exhibit positive deviation from ideal behaviour.
 (ii) Solutions having large positive deviation are constant boiling.
 (iii) The formation of solutions exhibiting positive deviation is accompanied with absorption of heat.
 (iv) Liquid pairs showing positive deviation have similar polarity.
 A i, ii ~~B i, iii~~
 C ii, iv ~~D ii, iii~~
- Q 54. Consider a galvanic cell having a potential of 3.17V and select the incorrect statement(s) from the following:
 (i) If an external opposite potential greater than 3.17V is applied to it, the spontaneous reaction will continue to take place.
 (ii) If an external opposite potential of 3.17V is applied to it, the non-spontaneous reaction will take place.
 A i ~~B ii~~
 C Both i and ii ~~D Neither i nor ii~~
- Q 55. For the reaction:
 $\text{A}^{2+} + \text{B}^+ \rightarrow \text{A}^{3+} + \text{B}$
 Calculate the maximum useful work done at 298K. Given: $K_c = 10^{10}$; $F = 96487 \text{ C/mol}$
 A -56927.3 J/mol ~~B +56927.3 J/mol~~
 C -0.59V ~~D +0.59V~~
- Q 56. During the process of charging of lead storage battery, the change in oxidation state of lead at the anode is:
 A +2 to 0 ~~B +4 to +2~~
 C +2 to +4 ~~D 0 to +2~~
- Q 57. Given below is a statement (S) and its reason (R):
 (S): A catalyst catalyses both spontaneous and non-spontaneous reactions.
 (R): A catalyst does not alter Gibb's free energy of a reaction.
 Select the correct choice:
 A Both S and R are correct and R is the explanation for S. ~~B Both S and R are correct but R is not the explanation for S~~
 C S is incorrect and R is correct ~~D S is correct and R is incorrect~~

Q 58. The equation for catalytic decomposition of phosphine is given below:



The molecularity of the reaction would be:

- ☒ A 4
☐ C 11

- ☐ B 1
☐ D None of the above

Q 59. The half-life of a first order reaction having rate constant of 200s^{-1} is:

- ☒ A $3.46 \times 10^3\text{s}$
☐ C $3.46 \times 10^{-2}\text{s}$

- ☒ B $3.46 \times 10^{-3}\text{s}$
☐ D 3.46s

Q 60. Select the correct statement(s) from the following:

- (i) Lanthanoids comprise of fourteen elements.
(ii) The first element of Lanthanoid series is Lanthanum
(iii) The symbol for Lanthanum is Ln.
(iv) All lanthanoids form stable tripositive ions.

- ☒ A ii, iii
☐ C i, ii

- ☐ B i, iv
☐ D ii, iv

Q 61. Give the magnetic properties of manganate and permanganate ions, respectively.

- ☒ A Both paramagnetic
☐ C Paramagnetic, diamagnetic

- ☐ B Both diamagnetic
☐ D Diamagnetic, paramagnetic

Q 62. Given below is a statement (S) and it's reason (R):

(S): $\text{Cu}^{2+}(\text{aq})$ is more stable than $\text{Cu}^{+}(\text{aq})$

(R): $\text{Cu}^{2+}(\text{aq})$ has a more negative enthalpy of hydration as compared to $\text{Cu}^{+}(\text{aq})$

Select the correct choice from the following:

- ☒ A Both S and R are correct and R is the explanation for S
☐ C S is incorrect and R is correct

- ☐ B Both S and R are correct but R is not the explanation for S
☐ D S is correct and R is incorrect

Q 63. The IUPAC name for $[\text{Pt}(\text{NH}_3)\text{BrCl}(\text{NO}_2)]^{-1}$ is:

- ☒ A Aminebromidechloridonitrito-N-platinate(II)
☐ C Amminebromidochloridonitrito-N-platinum(II)

- ☐ B Aminebromidochloridonitrite-N-platinum(II)
☐ D Amminebromidochloridonitrito-N-Platinate(II)

Q 64. $[\text{Ni}(\text{CO})_4]$ has tetrahedral geometry. Predict it's magnetic behaviour.
(Given: atomic number of Ni is 28)

- ☐ A Paramagnetic due to 2 electrons
☐ C Diamagnetic

- ☒ B Paramagnetic due to 4 electrons
☐ D None of the above

Q 65. In a tetrahedral crystal field, the energy of the e_g orbitals is:

- ☐ A Raised by $0.6\Delta_t$
☐ C Lowered by $0.6\Delta_t$

- ☒ B Raised by $0.4\Delta_t$
☐ D Lowered by $0.4\Delta_t$

Q 66. Methyl bromide and *tert*-butyl bromide are both made to undergo $\text{S}_\text{N}2$ reaction with aqueous NaOH separately. Which one of two will react faster?

- ☐ A *tert*-butyl bromide
☒ C Both will react at the same rate

- ☒ B Methyl bromide
☐ D Neither of the two will react

Q 67. In test tube 1 and 2, 1mL each of *n*-butyl chloride is taken. Next, 1mL ethyl alcohol is added to test tube 1 and 1mL water is added to test tube 2. Then small amount of NaOH is added to both 1 and 2. After mixing the contents, Br_2 in CCl_4 is added to both 1 and 2. Select the correct observations as given below:

- ☐ A 1 and 2 show orange colour
☐ C 1 is colourless and 2 has orange colour

- ☒ B 1 and 2 are colourless
☐ D 1 has orange colour and 2 is colourless

- Q 68. Given below is a statement (S) and it's reason (R):
 (S): Chlorine is an ortho/para- directing group
 (R): Chlorine is an electron withdrawing group
 Select the correct choice as given below:
 A Both S and R are correct and R is the explanation for S. ~~B Both S and R are correct but R is not the explanation for S~~
 C S is incorrect and R is correct ~~D S is correct and R is incorrect~~

- Q 69. 1-Methyleyclopentane-1,2-diol has which of the following functional groups?
~~A 2 primary -OH~~ B 2 secondary -OH
 C 1 primary -OH and 1 secondary -OH D 1 secondary -OH and 1 tertiary -OH

- Q 70. Compound A (C_4H_8) is reacted with water in presence of few drops of concentrated sulphuric acid. To the product obtained is added an equimolar mixture of anhydrous zinc chloride and concentrated hydrochloric acid. The reaction mixture instantly turns turbid. Compound A is most likely which one of the following?
~~A But-1-ene~~ B But-2-ene
 C Butan-2-ol D 2-methylpropene

- Q 71. In a mixture of ethyl chloride, ethyl alcohol and sodium hydroxide a reaction takes place. Select the correct statement(s) regarding the same from the following:
 (i) The reaction proceeds by S_N2 mechanism
 (ii) Ethyl alcohol is the attacking nucleophile
 (iii) Ethene is obtained as the product
 (iv) Sodium hydroxide is needed only in catalytic amounts
~~A i~~ B iv
 C ii, iii D iii, iv

- Q 72. The correct number and type of bonds present in $HCHO$ are:
~~A 2 sp^2 -s~~ B 2 sp^2 -s
 1 sp^2 - sp^3 1 sp^2 - sp^2
 1 p- sp^3 1 p-p
 C 2 sp^3 -s D 2 sp^3 -s
 1 sp^3 - sp^3 1 sp^3 - sp^2
 1 p-p 1 sp^2 - sp^2

- Q 73. Given below is a statement (S) and it's reason (R):
 (S): Ethers have lower boiling points than carbonyls of comparable masses
 (R): Ethers form stronger dipole - dipole interactions as compared to carbonyl compounds
 A Both S and R are correct and R is the explanation for S. B Both S and R are correct but R is not the explanation for S
~~C S is incorrect and R is correct~~ D S is correct and R is incorrect

- Q 74. Methyl bromide is reacted with Mg in dry ether solvent at $0^\circ C$. CO_2 gas is then passed through the resulting solution followed by dilute aqueous HCl. The product obtained will give which of the following test positive?
 A Lucas test B DNP test
~~C Sodium bicarbonate test~~ D None of the above

- Q 75. Methyl chloride upon reaction with $AgCN$ followed by hydrolysis gives compound X. What will be the molecular formula of X?
 A CH_3N B $C_2H_4O_2$
~~C Both A and B~~ D Neither A nor B

SERIES - B

- Q 76. What is the mass of 0.50 mol cobalt (II) chloride?
(Given: at mass of Co is 58.9u, at mass of Cl is 35.5u)
A 129.9g B 65g
☒ C 94.4g D 47.2g
- Q 77. If you had 50 mL of 0.01M HCl solution that could be exactly neutralized with 25 mL of a given NaOH solution, then what would be the concentration of this solution?
A 0.02M B 0.02 g/mL
☒ C 0.002 g/L D 0.002M
- Q 78. In the Antarctic cores, ice from glacial periods is found to be enriched with lighter H_2O^{16} as compared to heavier H_2O^{18} . Suggest a suitable reason for the same.
A H_2O^{16} and H_2O^{18} have different molecular mass B H_2O^{16} and H_2O^{18} have different boiling points
☒ C Both A and B ☒ D Neither A nor B
- Q 79. In Rutherford's α -particle scattering experiment, if the structure of the atom was as suggested previously by Thomson's model, which of the following statement(s) would be correct?
A Just a few α -particles would pass through without slowing down or deflecting by small angles. B All α -particles would pass through without slowing down or deflecting by small angles.
☒ C Just a few α -particles would slow down and deflect by small angles. ☒ D All the α -particles would slow down and deflect by small angles.
- Q 80. During a study of atomic and ionic radii of four species: (1) Na, (2) Na^+ , (3) F and (4) F^- four values, in pm, got mixed up These were : (i) 64, (ii) 95, (iii) 136 and (iv) 186. From the given below choices, select the pairing most likely to be correct.
A (1)-(iv), (3)-(i) B (3)-(i), (4)-(iii)
☒ C (1)-(i), (3)-(iv) D (2)-(iv), (3)-(ii)
- Q 81. Select the correct statement(s) from the following:
(a) Electronegativity is constant for an element
(b) Electronegativity is a measurable quantity
(c) Pauling scale is a numerical scale for electronegativity
(d) In a period, electronegativity increases from left to right
A (b), (c) B (c), (d)
☒ C (d) D all correct
- Q 82. In the stable shape of ClF_3 , as per VSEPR theory, the position of the lone-pairs is:
☒ A Only axial B Only equatorial
C Both axial and equatorial D Neither axial nor equatorial
- Q 83. Given below is a statement (S) and it's reason (R):
(S): As per MO theory, a bonding MO possesses energy higher than either of the two combining atomic orbitals
(R): Bonding MO does not have a nodal plane between the bonding nuclei
☒ A Both S and R are correct and R is the explanation for S B Both S and R are correct but R is not the explanation for S
C S is incorrect and R is correct D S is correct but R is incorrect
- Q 84. Select the incorrect statement(s) from the following:
(a) Enthalpy of fusion maybe negative or positive.
(b) During phase transformation at constant pressure, temperature slowly decreases.
(c) Standard enthalpy of sublimation is the change in enthalpy when one mole of a solid substance sublimates upon heating at standard pressure of 1 atm.
(d) Enthalpy is an intensive quantity.
A a,b,c,d B a,b,c
☒ C b,c,d D a,c,d

- Q 85. Upon adding sulphuric acid to 1L of water, the temperature is observed to rise by 17°C . Calculate the thermal energy lost from the mixture. (specific heat of water is $4.18\text{Jg}^{-1}\text{^{\circ}C}^{-1}$). (take density of water to be 1g/mL)
- A 71.06 kJ
C 71.06 J
B 17.1 kJ
D 17.1 J
- Q 86. In an aqueous solution of acetic acid, the concentration of hydrogen ions can be reduced by which of the following:
- A Addition of acetate ions
C Both A and B
B Addition of hydrogen ions from an external source
D Neither A nor B
- Q 87. The ionic product of water at 298K is:
- A $1 \times 10^{-7}\text{M}$
C $1 \times 10^{-14}\text{M}^2$
B $1 \times 10^7\text{M}$
D $1 \times 10^{14}\text{M}^2$
- Q 88. In $\text{Al}(\text{IO}_3)_3$, the oxidation state of Al, I and O, respectively, is:
- A -3, -5, -2
C +3, -5, -2
B +3, +5, +2
D +3, +5, -2
- Q 89. Among X_2 and HX ($\text{X}=\text{F}, \text{Cl}, \text{Br}, \text{I}$) the strongest oxidant and the strongest reductant, respectively, is:
- A F_2 , HF
C I_2 , HF
B F_2 , HI
D I_2 , HI
- Q 90. A student incorrectly suggested the IUPAC name of an alkane as 4-iso-propyl-6-secbutyl-nonane. Select the correct name for the same.
- A 4-isopropyl-6-secbutyl-nonane
C 4-sec-butyl-6-isopropyl-nonane
B 4-sec-butyl-6-iso-propyl-nonane
D Suggested name is correct

BIOLOGY (Q. No. 91 to Q. No. 135.)
(For Candidates Opted for PCB / PCBM Stream)

- Q 91. Match the following columns and select the correct option:-
- | Column I | Column II |
|----------------------|---|
| a. Medulla oblongata | 1. Responsible for cognitive functions like memory, communication and reasoning |
| b. Cerebrum | 2. Connects left and right cerebral hemispheres |
| c. Cerebellum | 3. Controls respiration and cardiovascular reflexes |
| d. Corpus callosum | 4. Maintains balance, posture and co-ordinates muscular movement |
- A a-3, b-1, c-4, d-2
C a-3, b-2, c-4, d-1
B a-2, b-4, c-1, d-3
D a-4, b-1, c-3, d-2
- Q 92. Animals characterized by unsegmented body, presence of water vascular system and endoskeleton of dermal calcareous plate belong to
- A Arthropoda
C Echinodermata
B Mollusca
D None of these
- Q 93. Which of the statements given below is/are correct?
- Statement 1: Hemis National Park is located in the cold desert of Ladakh.
Statement 2: Hemis National Park is also known as the 'Snow Leopard Capital of India'.
- A Only 1
C Both 1 and 2
B Only 2
D Neither 1 nor 2
- Q 94. The primary site of erythropoiesis is
- A Bone marrow
C Nerve cells
B Liver
D Mast cells

Q 95. Match the following columns and select the correct option:-

Column I

- a. Malleus, Incus, Stapes
- b. Fibula, Tibia, Femur
- c. Maxilla, Mandible, Hyoid
- d. Ulna, Humerus, Radius

Column II

- 1. Skull
- 2. Fore limb
- 3. Hind limb
- 4. Ear ossicles

- A a-4, b-2, c-1, d-3
~~B a-4, b-3, c-1, d-2~~

- B a-1, b-3, c-4, d-2
~~C a-1, b-2, c-4, d-3~~

Q 96. Remnants of nucellus in a mature seed is called

- ~~A Pericarp~~
~~C Perisperm~~

- B Periderm
~~D Periblem~~

Q 97. A PCR involves three steps repeated cyclically for DNA amplification. The correct order of these steps is

- ~~A Annealing → Extension → Denaturation~~

- B Extension → Annealing → Denaturation

- C Denaturation → Extension → Annealing

- ~~D Denaturation → Annealing → Extension~~

Q 98. In flowering plants, a seed develops from X and a fruit develops from Y

- A X - fertilized egg, Y - ovule

- B X - fertilized egg, Y - ovary

- ~~C X - ovule, Y - ovary~~

- ~~D X - ovary, Y - ovule~~

Q 99. What is true about PEN for angiosperms?

- A It is a diploid pre-fertilization structure

- B It is formed during double fertilization

- ~~C It is a triploid dormant nucleus that do not divide~~

- ~~D It is found in antipodal cells~~

Q 100. IVF, a type of assisted reproductive technology involves

- ~~A Transfer of sperms into the fallopian tube~~

- B Introduction of eggs into the fallopian tube

- C Transfer of embryos into the uterus

- ~~D None of these~~

Q 101. Which of the following sets of diseases is caused by viruses?

- ~~A Typhoid and Amoebiasis~~

- B Malaria and Filariasis

- C Hepatitis B and Typhoid

- ~~D Common Cold and Hepatitis B~~

Q 102. Which of the following pairs is not a method of in situ conservation of biodiversity?

- 1 Wildlife sanctuary and national park

- 2 Biosphere reserve and sacred grove

- ~~3 Botanical garden and zoological park~~

- 4 Arboretum and aquarium

- A 1 and 2 only

- B 2 and 3 only

- ~~C 3 and 4 only~~

- ~~D 2 and 4 only~~

Q 103. The increase in concentration of nutrients in a water body leading to excessive algae growth is called

- ~~A Biochemical oxygen demand~~

- B Chemical oxygen demand

- C Biomagnification

- ~~D Eutrophication~~

Q 104. The process by which a bacteriophage transfers genetic material from one bacterium to another is

- A Transduction

- ~~B Transcription~~

- C Translation

- ~~D Transformation~~

- Q 105. ADA deficiency is X genetic disorder that affects Y
 A X - an autosomal dominant, Y - erythrocytes
 B X - an autosomal recessive, Y - lymphocytes
 C X - a sex-linked dominant, Y - erythrocytes
 D X - a sex-linked recessive, Y - lymphocytes

- Q 106. Which of the following is not a part of an operon?
 A Structural gene
 B Promotor gene
 C Regulator gene
 D Co-ordinator gene

- Q 107. Which one of the following is incorrectly matched?
 1 Francois Jacob and Jacques Monod - Operon model of gene regulation
 2 James Watson and Francis Crick - Structure of DNA
 3 Mathew Meselson and F. Stahl - Replication of DNA
 4 Alfred Hershey and Martha Chase - DNA is the genetic material
 A 1 only
 B 1 and 3 only
 C 3 and 4 only
 D None of these

- Q 108. According to Hardy-Weinberg principle, allele and genotype frequencies in a population will remain constant in the presence of
 A Mutation
 B Genetic drift
 C Random mating
 D Gene flow

- Q 109. Colostrum, the yellowish fluid secreted by mother during initial days of lactation contains X that provide Y immunity to the infant
 A X - Immunoglobulins, Y - Active
 B X - Immunoglobulins, Y - Passive
 C X - Macrophages, Y - Auto
 D X - Macrophages, Y - Acquired

- Q 110. Which of the following statements are true?
 1 Plasmids have autonomous mode of replication
 2 Most plasmids are small circular DNA molecules
 3 Some plasmids are present as only 1-2 copies per cell, whereas the others may have 20-100 copies per cell
 4 Many plasmids carry genes that confer antibiotic resistance
 A 1, 2 and 4 only
 B 2 and 4 only
 C 1 and 2 only
 D 1, 2, 3 and 4

- Q 111. The unauthorized exploitation of bio-resources and traditional knowledge of countries and people concerned by multi-national companies and other organizations, often without any compensation is termed as
 A Biopatenting
 B Bioprospecting
 C Bioethics
 D Biopiracy

- Q 112. In a dihybrid cross $AAbb \times aaBB$ governed by Mendelian inheritance, if the offspring is made to cross with homozygous recessive for both the genes, what will be the genotypic and phenotypic ratio of the progeny obtained?
 A 9:3:3:1 and 12:3:1, respectively
 B 1:2:1 and 1:2:1, respectively
 C 1:1:1:1 and 1:1:1:1, respectively
 D 3:1 and 9:3:4, respectively

- Q 113. If a color-blind man marries a woman who is homozygous for normal colour vision, the probability of their son and daughter being colour-blind is
 A Son - 0.5, Daughter - 0.5
 B Son - 1, Daughter - 0.5
 C Son - 1, Daughter - 1
 D Son - 0, Daughter - 0

- Q 114. Origin of homologous organs is due to
 A Divergent evolution
 B Convergent evolution
 C Adaptive radiation
 D Mutation

- Q 115. Cocaine affects the central nervous system and interferes with the transport of
☒ A Acetylcholine ☐ B Adrenaline
☐ C Histamine ☐ D Dopamine
- Q 116. Beekeeping is synonymous with
☒ A Pisciculture ☒ B Apiculture
☒ C Sericulture ☐ D Aquaculture
- Q 117. Which of the following statement is incorrect?
☒ A Survival of sperm depends on the pH of the medium and is more active in an alkaline medium
☐ B Menstrual flow occurs due to lack of progesterone
☒ C Site of fertilization in a mammal is the uterus ☒ D Seminal fluid is rich in fructose that provides energy for the spermatozoa
- Q 118. The first organisms to inhabit a bare rock in an area would be
☐ A Green algae ☐ B Mosses
☐ C Fungi ☒ D Lichens
- Q 119. Which of the following are free-living nitrogen fixing bacteria
 1 Rhizobium
 2 Azospirillum
 3 Anabaena
 4 Azotobacter
☒ A 1 and 3 only ☐ B 2 and 4 only
☒ C 3 and 4 only ☐ D 2 and 3 only
- Q 120. Which of the following is common to asexual and sexual reproduction?
☐ A Transfer of genetic material ☐ B Reduction division
☒ C Gametic fusion ☐ D Universal occurrence in all living organisms
- Q 121. Which of the following taxonomical aids serve as a referral system for identification of both plants and animals?
☒ A Herbaria and museum ☐ B Museum and keys
☐ C Herbaria and keys ☐ D Herbaria, museum and keys
- Q 122. Which of the following statements is incorrect?
☒ A Viroids are infectious particles that are devoid of a protein coat
☒ B Presence of a cell wall is a feature of members of Kingdom Fungi and Plantae
☒ C Kingdom Mnpera and Protista include diverse prokaryotic life forms
☒ D Kingdom Animalia consists of eukaryotic and heterotrophic organisms
- Q 123. The spore bearing leaf in Pteridophytes is called a
☐ A Prothallus ☐ B Protonema
☐ C Sorus ☒ D Sporophyll
- Q 124. Select the correct pair of character present in the respective angiosperm family
☐ A Solanaceae: Tricarpellary, syncarpous ☒ B Fabaceae: Monocarpellary, unilocular ovary with marginal placentation
☐ C Solanaceae: Bilocular, inferior ovary with swollen placenta ☐ D Liliaceae: Bilocular, inferior ovary with axile placentation

Q 125.

Match the following columns and select the correct option:-

Column I

a. Bulliform cells

b. Idioblasts

c. Companion cells

d. Sclereids

Column II

1. Specialized parenchymatous cells associated with sieve tube elements

2. Large, colourless motor cells present in leaves of grasses

3. Dead cells with highly thickened cell wall commonly found in the hard parts of the plant

4. Parenchymatous cells that store waste materials, pigments and minerals

A a-3, b-2, c-4, d-1

☒ E a-2, b-4, c-1, d-3

B a-4, b-2, c-1, d-3

D a-2, b-4, c-3, d-1

Q 126.

If a plasmolyzed cell is placed in a hypotonic solution, then

A Exosmosis takes place and cell shows contraction of protoplast

☒ E Neither exosmosis nor endosmosis takes place and cell becomes flaccid

B Endosmosis takes place and cell becomes turgid

D None of these

Q 127.

Which of the following options is correct for the following two statements:

Statement 1: The water potential of pure water at standard temperatures, which is not under any pressure, is taken to be zero.

Statement 2: The two main components that determine water potential are solute potential and pressure potential.

A Both the statements are true

B Both the statements are false

C Statement 1 is true while statement 2 is false

☒ D Statement 1 is false while statement 2 is true

Q 128.

Formation of interfascicular cambium and cork cambium from parenchyma cells is an example of

A Differentiation

B Dedifferentiation

☒ C Redifferentiation

D Photoperiodism

Q 129.

CO₂ is released during

A Aerobic respiration and alcoholic fermentation

C Alcoholic fermentation and lactic acid fermentation

☒ D Aerobic respiration, alcoholic fermentation and lactic acid fermentation

Q 130.

Which element is involved in splitting of water to release oxygen during photosynthesis?

☒ A Manganese

B Molybdenum

C Copper

D Boron

Q 131.

Disjunction of homologous chromosomes occurs in which phase of meiosis?

A Anaphase I

☒ B Anaphase II

C Telophase I

D Telophase II

Q 132.

Which of the following statements is correct with respect to cockroaches?

A Sperms are stored in vas deferens in males

C Ovaries are found lying in 2nd - 6th abdominal segments

☒ B Malpighian tubules absorb nitrogenous waste products and convert them into ammonia

D Tegmina, the forewings, are primarily used for flight

Q 133.

In human beings, ciliated epithelium is present in

1 Respiratory tract

☒ 2 Fallopian tubes

3 Gastro intestinal tract

4 Pancreatic duct

A 1 and 2 only

☒ C 1, 2 and 3 only

B 1 and 3 only

D 1 and 4 only

- Q 134. Golgi complex is involved in
 A Amino acid activation and protein digestion
 B Post-translational modification of proteins
 C Breakdown of fatty acids
 D Hydrolysis of carbohydrates
- Q 135. A structural polysaccharide found in fungal cell wall is
 A Lecithin
 B Chitin
 C Pectin
 D Glycogen

Mathematics (Q. No. 136 to Q. No. 180.)
(For Candidates Opted for PCM / PCBM Stream)

- Q 136. If two events A and B are such that $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$, what is $P(\bar{A} \text{ and } \bar{B})$?
 A 0.3
 B 0.9
 C 0.7
 D 0.1
- Q 137. Let $f(x) = \sin x$, $g(x) = x^2$ and $h(x) = \log_e x$. If $F(x) = (hogof)(x)$, then $F'(x)$ is equal to
 A $-2 \operatorname{cosec}^2 x$
 B $2 \cot x$
 C $2 \log_e \sin x$
 D $\operatorname{cosec}^2 x$
- Q 138. For a function $f(x)$, if the tangent at $x = a$ is vertical, then:
 A $f'(a) = 0$
 B $f(a) = 0$
 C $f'(a)$ does not exist
 D The function is not defined at $x = a$
- Q 139. The mean of a set of numbers is 6. If each number is multiplied by 3, then the mean of new set is
 A 54
 B 12
 C 6
 D 18
- Q 140. A green dice and a red dice are thrown at random then the probability that number on red dice is greater than number on green dice is:
 A $1/2$
 B $5/12$
 C $5/9$
 D $5/6$
- Q 141. Which of the following relations on R is an equivalence relation?
 A $a R_1 b \Leftrightarrow |a| = |b|$
 B $a R_2 b \Leftrightarrow a \geq b$
 C $a R_3 b \Leftrightarrow a \text{ divides } b$
 D $a R_4 b \Leftrightarrow a < b$
- Q 142. The range of the function $f(x) = \frac{3}{2-x^2}$ is
 A $R - \{-\sqrt{2}, \sqrt{2}\}$
 B $(-\infty, 0) \cup [\sqrt{2}, \infty)$
 C $(-\infty, 0) \cup (0, \infty)$
 D $(-\infty, 0) \cup [2, \infty)$